

JD Sports

Interview Presentation

—

16 September 2023

Understand the problem

The CRM team would like to send email campaigns with the next best brand recommendation to customers who are most likely to repurchase in the next 3 months.

Tasks to complete

Build a model based on the given dataset and identify which customers can be targeted.

If each email costs the company £2, what's the total misclassification cost?

Understand the data: Variable Definition (17 columns, 101000 records or rows)

- **customer_site_id (str)** = Customer Identifier
- **orders_1yr (int)** = Number of orders placed in last 12 months
- **line_items_1yr (int)** = No. of items purchased in last 12 months
- **days_since_last_purchase_1yr (int)** = Days since last purchase in last 12 months
- **prop_reduced_items_1yr (float)** = Proportion of reduced priced items bought in the last 12 months
- **promo_customer_blc_1yr (int)** = whether the customer used blue light promotion offer in the last 12 months (0, 1)
- **promo_customer_freedelivery_1yr (int)** = whether free delivery was offered to the customer in the last 12 months (0, 1)
- **Brand_Nike_prop_revenue_1yr (float)** = Proportion of revenue from Nike product purchases in the last 12 months
- **Brand_Adidas_prop_revenue_1yr (float)** = Proportion of revenue from Adidas product purchases in the last 12 months
- **Device_mobile_prop_revenue_1yr (float)** = Proportion of revenue from purchases on a mobile device in the last 12 months
- **Device_iosapp_prop_revenue_1yr (float)** = Proportion of revenue from purchases on a ios device in the last 12 months
- **Delivery_NextDay_prop_revenue_1yr (float)** = Proportion of revenue from purchases with Next Day Delivery in the last 12 months
- **Payment_bnpl_prop_revenue_1yr (float)** = whether the customer used bnpl (buy now pay later) in the last 12 months
- **amount_paid_per_order_1yr (float)** = Total order amount before delivery
- **Current_RFM_Segment (str)** = RFM (Recency, Frequency, Monetary) segment of the customer
- **ONS_Segment (str)** = Socio-demographic segment for the customer (from ONS)
- **repurchased (int)** = Whether the customer made a repeat purchase in the next 3 months (0, 1)

Pre-processing and cleaning data

(Remove missing data in the record and map some string columns to make it comparable with other features for model training)

Data-Cleaning & Pre-processing

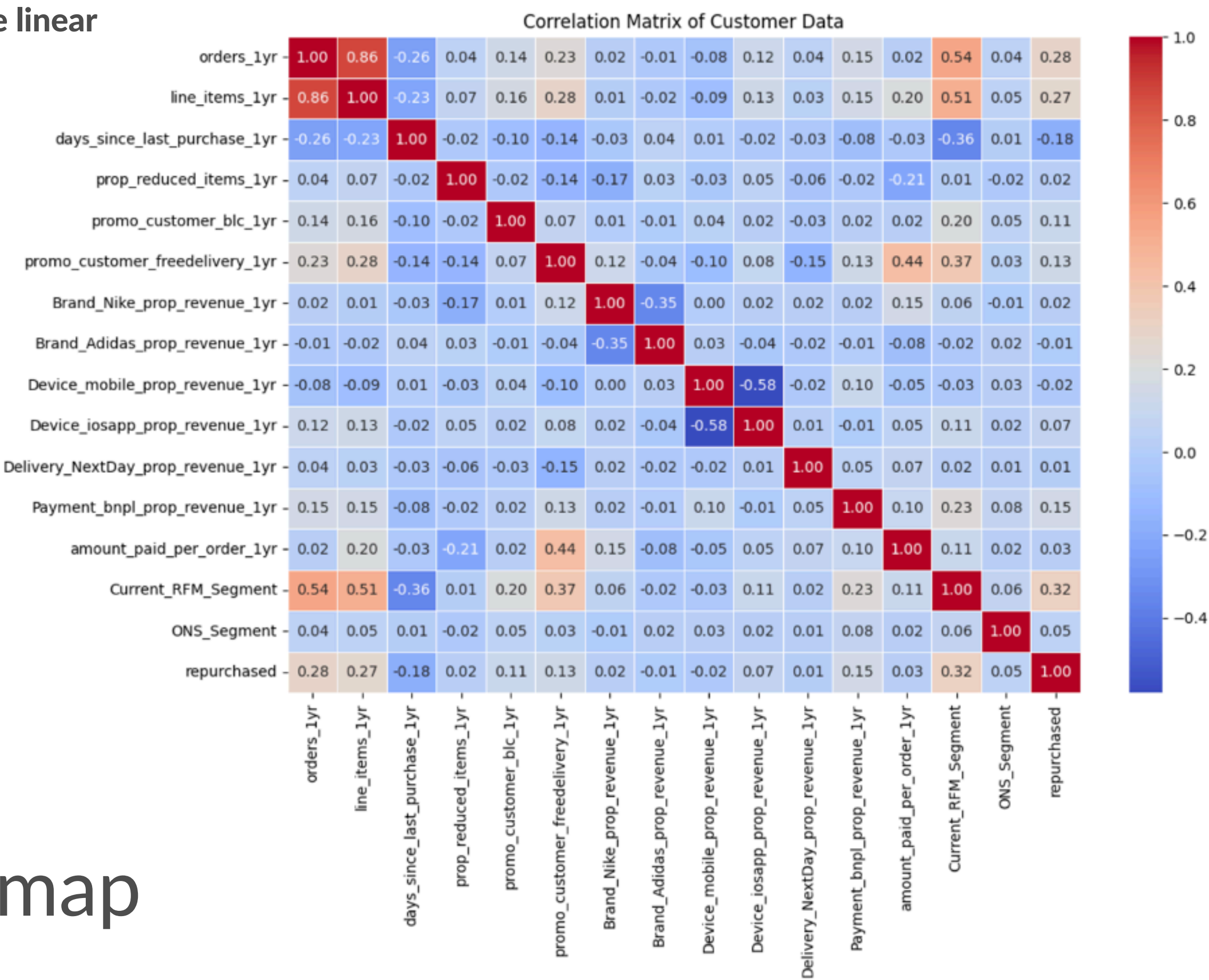
```
customer_site_id      0
orders_1yr            8
line_items_1yr        0
days_since_last_purchase_1yr  8
prop_reduced_items_1yr  0
promo_customer_blc_1yr  0
promo_customer_freedom_delivery_1yr  0
Brand_Nike_prop_revenue_1yr  0
Brand_Adidas_prop_revenue_1yr  0
Device_mobile_prop_revenue_1yr  0
Device_iosapp_prop_revenue_1yr  0
Delivery_NextDay_prop_revenue_1yr  0
Payment_bnpl_prop_revenue_1yr  0
amount_paid_per_order_1yr  0
Current_RFM_Segment   0
ONS_Segment           0
repurchased           0
```

101000 -> 100992 : only 8 rows has been removed in the process

- We chose to remove the missing values rather than filling them with the average or another number, as we have sufficient data, and imputation would introduce unnecessary and potentially misleading information
- Converted some string columns to categorical numbers to make them comparable with other features for model training

Use Pearson correlation heatmap to examine the linear relationships between the features in dataset.

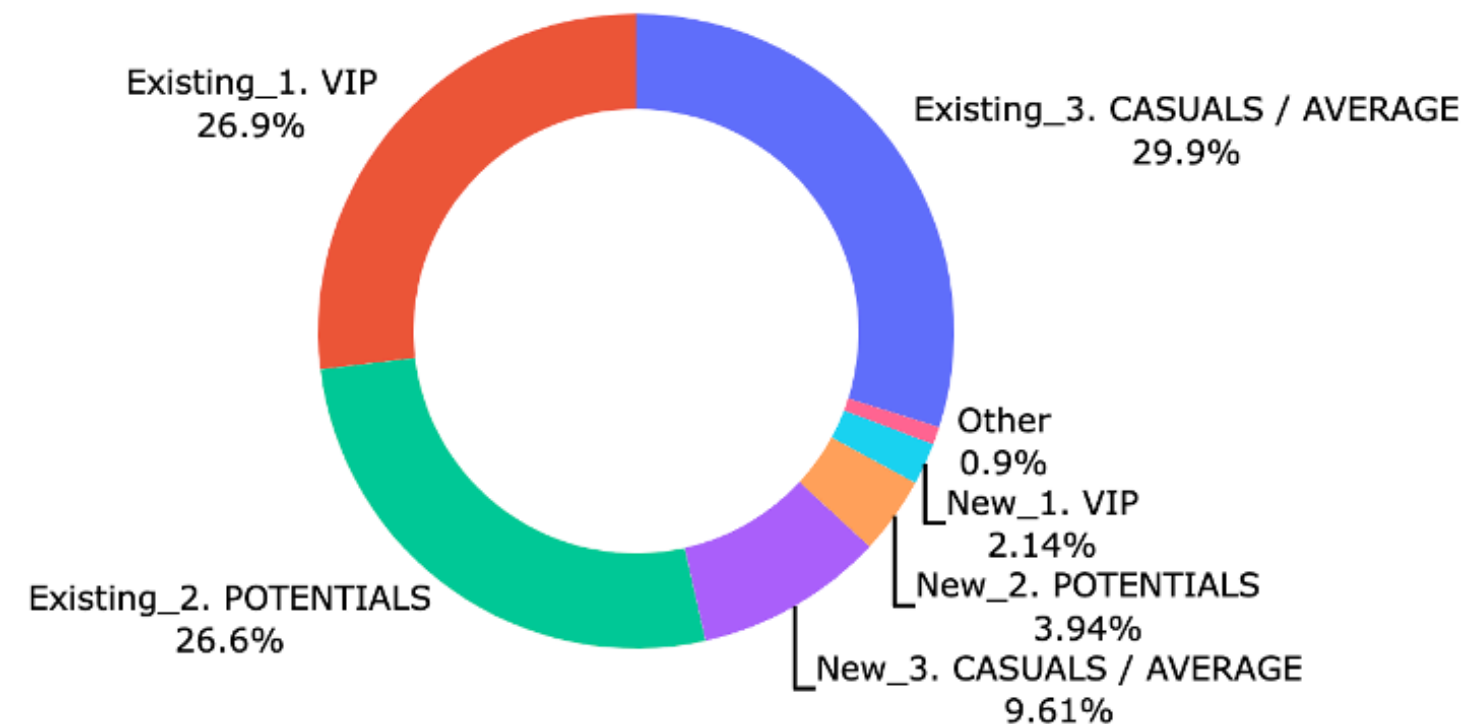
Key insights from the heatmap include a strong positive correlation between "repurchased" and "Current_RFM_Segment", as well as "orders_1yr" and line_item_1yr. These features appear to have a more significant relationship with customer repurchasing behaviour, providing valuable insights for predicting future customer activity.



Creative Presentation

Correlation Heatmap

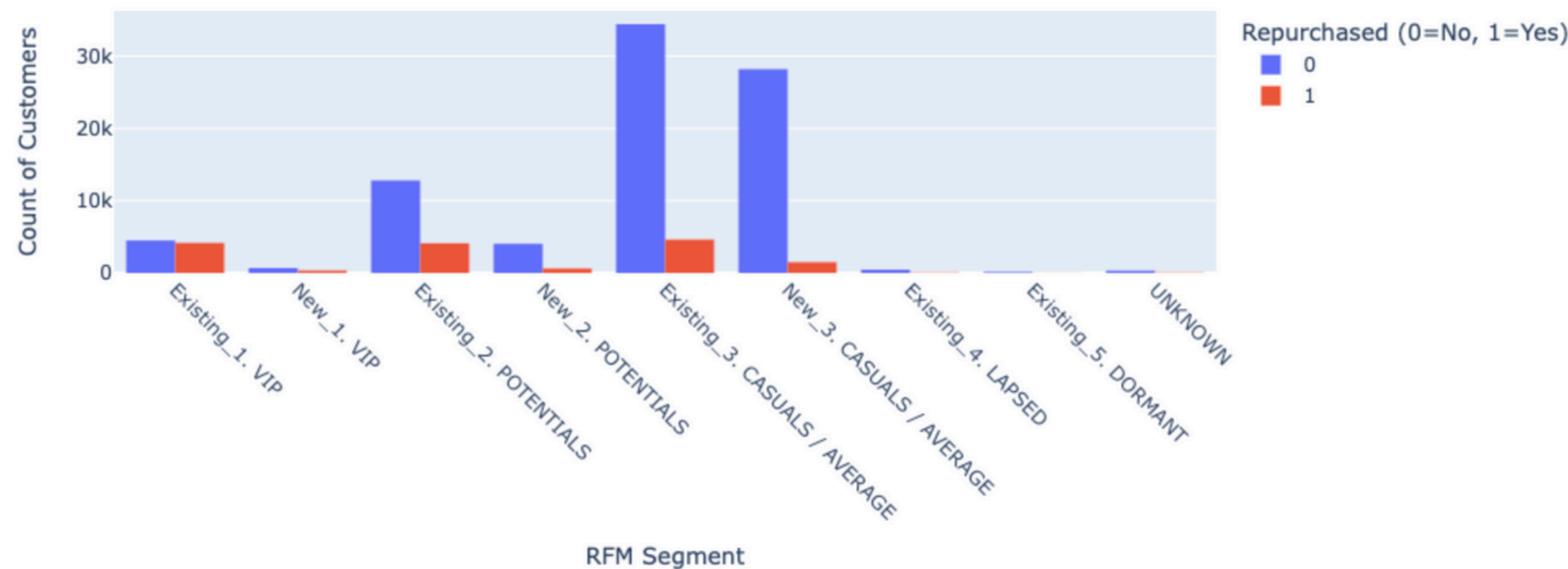
Distribution of Current RFM Segment for Repurchased Customers



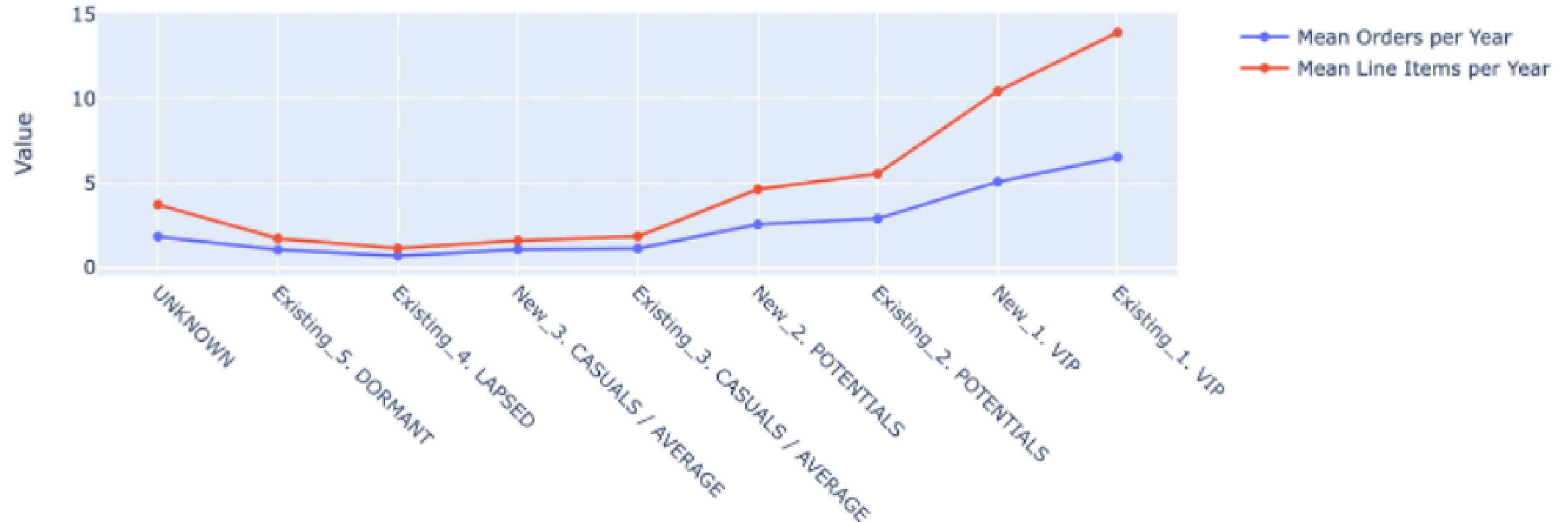
Existing customers in all categories have the highest probability of making repeat purchases within the next 3 months. Notably, the casual/average customers segment represents the largest share of repeat buyers, indicating that this group is highly engaged and likely to make subsequent purchases. Additionally, the new customers segment shows significant potential, with a strong likelihood of repurchasing shortly after their initial transaction. This analysis highlights the importance of nurturing relationships with both established and new customers to drive future sales.

To better understand why "Current_RFM_Segment" is the most influential feature in predicting repurchases, we use histogram along with pie chart that breaks down repurchase behaviour by customer segment.

Histogram of RFM Segments Categorized by Repurchase Status



RFM Segment Analysis



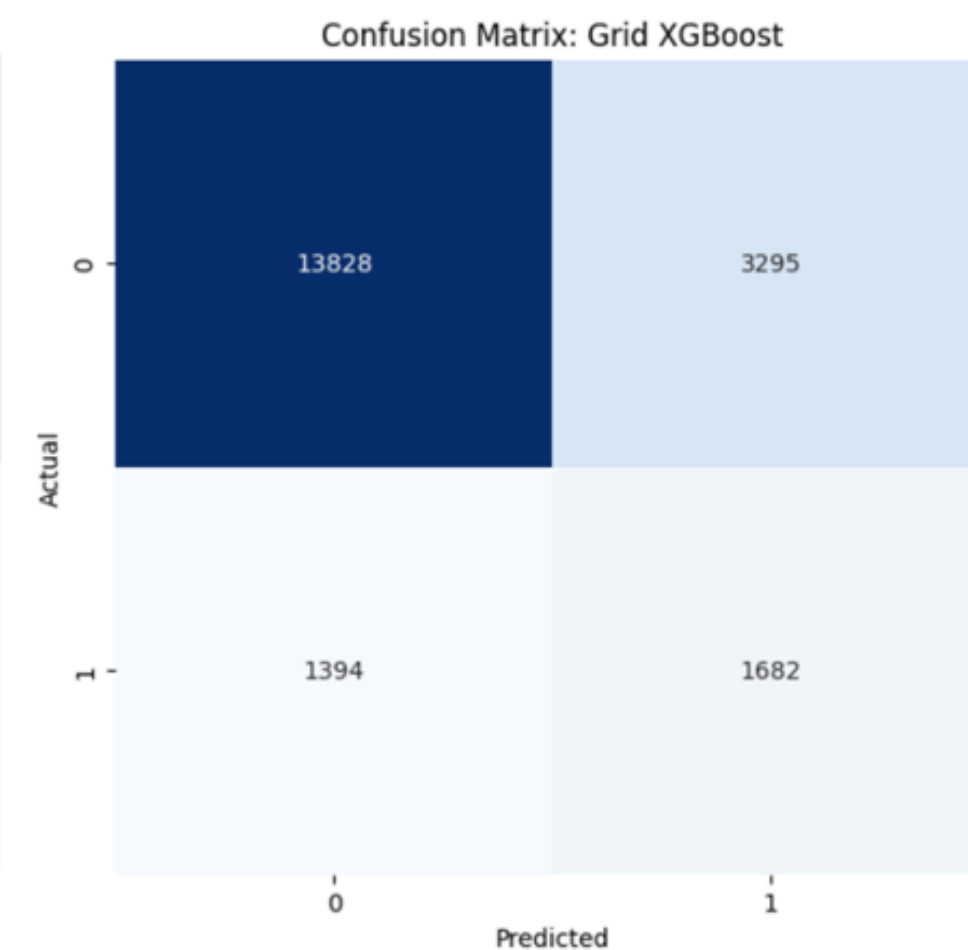
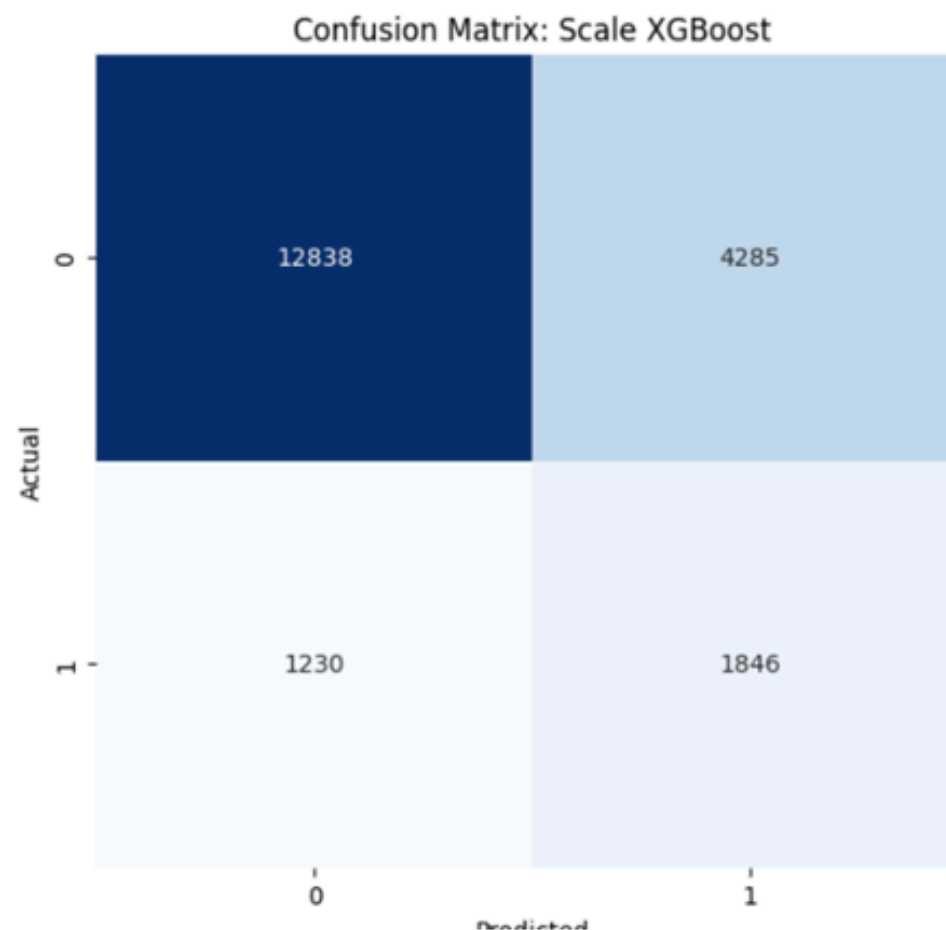
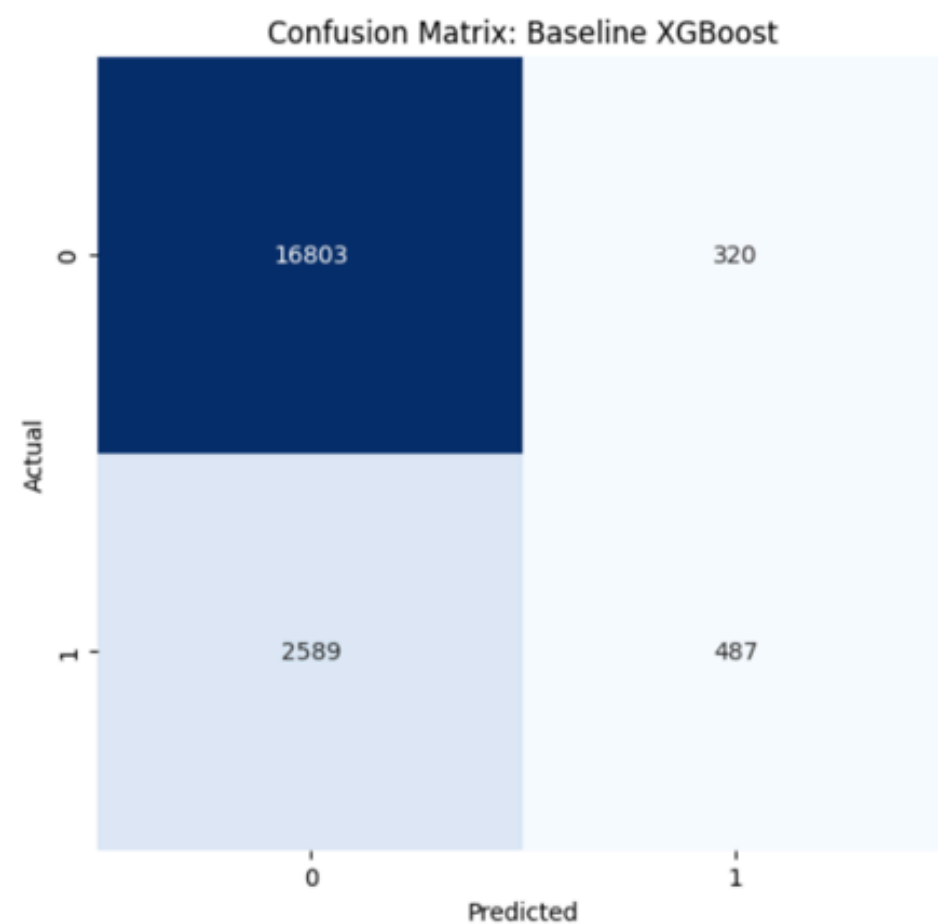
- The average line graph for both **orders per year** and **line items per year** highlights the **Existing** and **New VIP** groups stand out, placing an average of **10 orders per year** and **purchasing more than 5 items** annually.
- The **Potential group** records a higher average number of orders compared to Casual customers, while the Casual, Lapsed, and Dormant segments exhibit the lowest order and purchase activity throughout the year.
- Customers who **place more orders** tend to **purchase a higher** number of items. This suggests that order frequency is closely tied to the total number of items purchased, reinforcing the link between customer activity and engagement.

Model Training

We use machine learning models to identify customers who are most likely to repurchase within the next 3 months.

Model	0 (Not repurchase)		1 (repurchase)		Training time	Overall Accuracy
	Precision	Recall	Precision	Recall		
Baseline models without optimising hyperparameters and remove features						
Decision Tree	0.88	0.85	0.28	0.32	372 ms	0.77
Random Forest	0.87	0.98	0.59	0.19	5.57 s	0.86
XGBoost	0.87	0.98	0.59	0.18	983 ms	0.86
Discard uncorrelated features from the dataset						
XGBoost (Remove Features)	0.87	0.98	0.60	0.16	778 ms	0.86
Adjust parameter to solve the unbalance class problem lead over fitted model						
XGBoost (Remove Features)	0.91	0.75	0.30	0.60	285 ms	0.73
Fine-tunning model using Grid search to search the most optimal set of parameters						
XGBoost (Remove Features)	0.91	0.81	0.34	0.55	59min 2s	0.77

- XGBoost provides a faster training time when compare with the random forest which achieve the same accuracy
- Drop features down to 4 main features ['orders_1yr', 'line_items_1yr', 'days_since_last_purchase_1yr', 'Current_RFM_Segment']
- The scale_pos_weight parameter in XGBoost model can help balance the weight of the minority class relative to the majority class.
- Using GridSearchCV to explore all possible combinations of hyperparameters in XGBoost



model	TN	FN	TP	FP	Total email sends	Total cost	Misclassification cost
XGBoost-Baseline	16803	2589	487	320	807	1614 £	640 £
XGBoost-Scale	12838	1230	1846	4285	6131	12262 £	8570 £
XGBoost-Grid-search	13828	1349	1682	3295	4977	9954 £	6590 £
			Send the email to correct target	Send the email to incorrect target (misclassification)			

Conclusion

To determine the best model from the table, we need to balance both minimising misclassification cost (FP) and maximizing the number of True Positives (TP)

XGBoost-Baseline: Misclassification Cost per TP= 640 £/ 487 $\approx$ **1.31 £/TP**

XGBoost-Scale: Misclassification Cost per TP= 8570 £/ 1846 $\approx$ **4.64 £/TP**

XGBoost-Grid-search: Misclassification Cost per TP= 6590 £/ 1682 $\approx$ **3.92 £/TP**



If minimizing misclassification cost per TP is your primary goal, then XGBoost-Baseline is the best choice

XGBoost-Baseline



If achieving the highest number of Target customer is more important, XGBoost-Scale is preferable despite its higher misclassification cost per TP.

XGBoost-Scale



If prioritize both misclassification costs and maximise the target customer this model seems to achieve the prediction to find the target customer while preserving the cost of an email.

XGBoost-Grid-search

Creative Presentation

Thank You

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